

A Rydberg Atom-Based Mixer: Measuring the Phase of a Radio Frequency Wave

Matthew T. Simons,¹ Abdulaziz H. Haddab,² Joshua A. Gordon,³ and Christopher L. Holloway³

¹*Department of Physics, University of Colorado, Boulder, Colorado 80302, USA*^{a)}

²*Department of Electrical Engineering, University of Colorado, Boulder, Colorado 80302, USA*

³*National Institute of Standards and Technology (NIST), RF Technology Division, U.S. Department of Commerce, Boulder Laboratories, Boulder, Colorado 80305, USA*

(Dated: 9 March 2019)

Rydberg atoms have been shown to be very useful in performing absolute measurements of the magnitude of a radio frequency (RF) field using electromagnetically-induced transparency (EIT). However, there has been less success in using Rydberg atoms for the measurement of the phase of an RF field. Measuring the phase of a RF field is a necessary component for many important applications, including antenna metrology, communications, and radar. We demonstrate a scheme for measuring the phase of an RF field by using Rydberg atoms as a mixer to down-convert an RF field at 20 GHz to an intermediate frequency on the order of kHz. The phase of the intermediate frequency corresponds directly to the phase of the RF field. We use this approach to measure the phase shift on an electromagnetic wave from a horn antenna as the antenna is placed at different distances from the Rydberg atom sensor. The atom-based RF phase measurements allow us to measure the propagation constant of the RF wave to within 0.1 % of the theoretical value.

Recently, great progress has been achieved in the development of a spectroscopic approach for radio-frequency (RF) electric (E) field measurements^{1–7}. This approach utilizes electromagnetically induced transparency (EIT) and Autler-Townes (AT) splitting to create a direct International System of Units (SI) traceable, self-calibrated measurement of an RF E-field amplitude. Rydberg atoms are very sensitive to RF fields due to their large dipole moments over a broad frequency range, from MHz to THz^{5,8,9}. Thus far Rydberg atoms have been used to extend the frequency range and sensitivity of RF electric field amplitude^{3,10} and polarization measurements⁷, but have not been used for RF phase measurements. This Rydberg-atom based sensor can act as a compact receiver/antenna for detecting modulated signals, which allows for communications applications. The first demonstration of this was presented in¹¹, and later by others^{12–18}. This has led to the new term “atom-radio”^{15,16}. The work in Refs. ^{11–18} have demonstrated the detection and reception of amplitude modulated (AM) and frequency modulated (FM) signals, where the AM and FM detection is based on the measurement of the amplitude of the carrier. To detect and receive BPSK/QPSK/QAM digital communication signals requires the ability to measure the phase of an RF field. Other applications such as near-field antenna metrology and radar also require a phase measurement.

The motivation of this work is to advance quantum-based E-field sensor technology. To this end, we demonstrate the ability of Rydberg atoms to measure the phase of an RF field. Including phase measurements allows the Rydberg atom system to fully characterize an RF E-field in a single quantum-based sensor. This technique relies on using the atom system as a Rydberg atom-based mixer. Applying an RF field that is on-resonance with the Rydberg transition acts as a local oscillator (‘LO’ in Fig 1), which causes the EIT/AT effect in

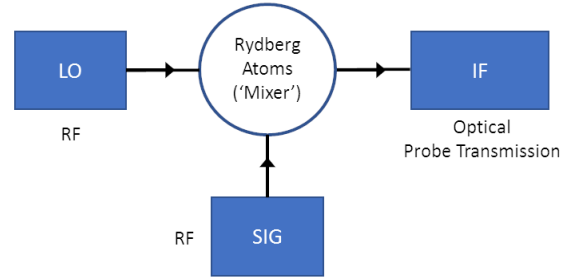


FIG. 1. Block diagram of the Rydberg atom ‘mixer’. The Rydberg atoms separate the difference frequency (IF) from two RF signals (LO and SIG). This demodulated signal is carried in the probe laser.

the Rydberg atoms to demodulate a second, co-polarized RF field (labeled ‘SIG’ in Fig 1). The difference frequency, or intermediate frequency (IF), is detected by optically probing the Rydberg atoms (Fig 1). The phase of the IF signal corresponds directly to the relative phase between the LO and RF signals. This technique can be applied over the same frequency range as E-field amplitude measurements, resulting in an atom-based sensor that can measure amplitude, polarization, and now phase for fields from 500 MHz to 1 THz in a single setup.

Rydberg atoms can detect and measure RF E-fields through the use of two related optical phenomena. EIT occurs when a laser field (probe field) that is on-resonance with an atomic transition would normally be absorbed by an atomic vapor, is instead transmitted in the presence of a second laser field (coupling field), which is tuned to another atomic transition. This occurs over a very narrow frequency range. When a third electromagnetic field, tuned to another linked atomic transition, is present, the original transparency region is split into two regions separated in frequency (AT-splitting). The frequency separation is directly related to the strength of the third RF field (typically an RF field). More detail on this process can

^{a)}Electronic mail: matthew.t.simons@colorado.edu.

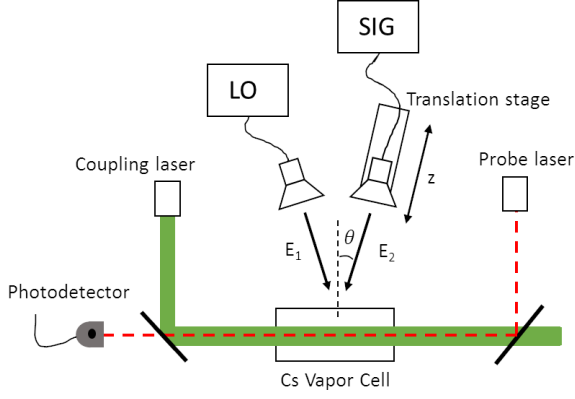


FIG. 2. A probe and coupling laser are overlapped through a Cs vapor cell using two dichroic mirrors. Two RF fields (E_1 and E_2) are applied using two horn antennas fed by two signal generators (LO and SIG). The SIG horn is placed on a micrometer translation stage. Transmission of the probe through the ^{133}Cs vapor is monitored by a balanced photodetector. Part of the probe beam is split and sent to the second port of the balanced photodetector without sampling the Cs (not pictured).

be found in ¹⁻⁶.

In this work, the atomic vapor cell is filled with cesium (^{133}Cs) atoms. Fig. 2 shows a diagram of the experimental setup. The probe laser is tuned to a wavelength of 852 nm to excite the ^{133}Cs from the ground state to the first excited state ($6S_{1/2} \rightarrow 6P_{3/2}$). The coupling laser is counter-propagated through the probe laser and tuned to 511.148 nm, coupling the excited state to the Rydberg state $34D_{5/2}$, and creating EIT in the probe laser, increasing the transmission through the vapor cell. The probe laser has a full-width at half-maximum (FWHM) of 425 μm and a power of 49 μW . The coupling laser has a FWHM of 620 μm and a power of 60.6 mW. Applying an RF field of 19.626 GHz causes a transition of $34D_{5/2} \rightarrow 35P_{3/2}$, producing AT-splitting.

With the probe laser frequency fixed at the first excited state transition frequency, the transmission through the vapor cell is reduced. The interference of two RF fields can then be detected by the effect on the EIT/AT splitting. By setting the probe laser frequency to the center of the EIT peak, the probe laser transmission is modulated by the interference between the two RF fields. If one of the RF fields is detuned from the other, the beat note can be observed in time in the transmission. This effect is demonstrated by placing two standard gain horn antennas 12 cm from the Cs vapor cell (Fig. 2). One antenna is fed by a signal generator set at $f_{\text{LO}} = 19.626000$ GHz to act as the LO. Another signal generator feeds the second RF signal, f_{SIG} . The two signal generators are connected to a rubidium (Rb) clock 10 MHz signal. Note that this synchronization signal need not be physically connected if each signal generator is connected to atomic clocks that were previously synchronized, allowing the LO and Rydberg atom system to be physically separated from the SIG signal.

The total electric field at the vapor cell ($\vec{E}_{\text{atoms}}$) is the

sum of the LO and SIG fields (E_1 and E_2). For two co-polarized fields, this is the scalar total $E_{\text{atoms}} = E_1 + E_2$. We define the LO and SIG fields as $E_1 = E_{\text{LO}} \cos(\omega_{\text{LO}}t + \phi_{\text{LO}})$ and $E_2 = E_{\text{SIG}} \cos(\omega_{\text{SIG}}t + \phi_{\text{SIG}})$. Assuming that the difference between the two frequencies $\Delta\omega = \omega_{\text{LO}} - \omega_{\text{SIG}}$ is much less than average frequency $\bar{\omega} = (\omega_{\text{LO}} + \omega_{\text{SIG}})/2$, such that $\Delta\omega \ll \bar{\omega}$, then we can express total field E_{atoms} as

$$E_{\text{atoms}} = E_1 + E_2 \quad (1)$$

$$= \cos(\omega_{\text{LO}}t + \phi_{\text{LO}}) \cdot$$

$$\sqrt{E_{\text{LO}}^2 + E_{\text{SIG}}^2 + 2E_{\text{LO}}E_{\text{SIG}} \cos(\Delta\omega t + \Delta\phi)} \quad (2)$$

where $\omega_{\text{LO}} = 2\pi f_{\text{LO}}$, $\omega_{\text{SIG}} = 2\pi f_{\text{SIG}}$, and $\Delta\phi = \phi_{\text{LO}} - \phi_{\text{SIG}}$. Equation 2 can be written to separate the term on resonance with the Rydberg transition $E_{\text{res}}(\omega_{\text{LO}})$ from the low frequency term $E_{\text{mod}}(\Delta\omega)$ as $E_{\text{atoms}} = E_{\text{res}} \cdot E_{\text{mod}}$, where

$$E_{\text{res}} = \cos(\omega_{\text{LO}}t + \phi_{\text{LO}}), \quad (3)$$

$$E_{\text{mod}} = \sqrt{E_{\text{LO}}^2 + E_{\text{SIG}}^2 + 2E_{\text{LO}}E_{\text{SIG}} \cos(\Delta\omega t + \Delta\phi)}. \quad (4)$$

The resonant term E_{res} causes an Autler-Townes effect as described above, reducing the peak of the EIT line. The low frequency term E_{mod} modulates the amplitude of the resonant effect.

The probe transmission is a function of the absorption of the atomic vapor ¹⁹:

$$T_{\text{probe}} \propto \exp\left(-\frac{2\pi L}{\lambda_p} \text{Im}[\chi]\right), \quad (5)$$

where L is the length of the cell, λ_p is the wavelength of the probe laser, and χ is the susceptibility. The susceptibility is a function of the E-field strength $|E_{\text{atoms}}|$ at ω_{LO} , thus the probe transmission is a function of the magnitude of the E-field on resonance with the Rydberg transition. The magnitude of the resonant term $|E_{\text{res}}| = 1$, so $|E_{\text{atoms}}| = |E_{\text{res}} \cdot E_{\text{mod}}| = |E_{\text{mod}}|$. For a SIG field that is small compared to the LO field ($E_{\text{SIG}} \ll E_{\text{LO}}$), the magnitude of the field can be written as

$$|E_{\text{atoms}}| \approx E_{\text{LO}} + E_{\text{SIG}} \cos(\Delta\omega t + \Delta\phi). \quad (6)$$

The atoms demodulate the high-frequency ω_{LO} field, and the probe transmission as a function of time is given by:

$$T_{\text{probe}} \propto |E_{\text{atoms}}| \approx E_{\text{LO}} + E_{\text{SIG}} \cos(\Delta\omega t + \Delta\phi). \quad (7)$$

The term E_{res} causes AT-splitting through the resonant interaction which reduces the probe transmission (T_{probe}) below the EIT peak. This reduction is then modulated in time by the low-frequency beat note, E_{mod} . In effect, the probe laser transmission varies in time by the difference frequency $\Delta\omega$, with a phase given by the difference between the LO and SIG phases $\Delta\phi$.

The observation of a signal at $\Delta f = \Delta\omega/2\pi$ in the detected probe transmission indicates the presence of an RF field at either $f_{\text{SIG}} = f_{\text{LO}} \pm \Delta f$. This is demonstrated in Fig. 3, where a discrete Fourier transform (DFT) of the probe signal is plotted as the SIG is varied from $f_{\text{SIG}} = 19.626001$ GHz to

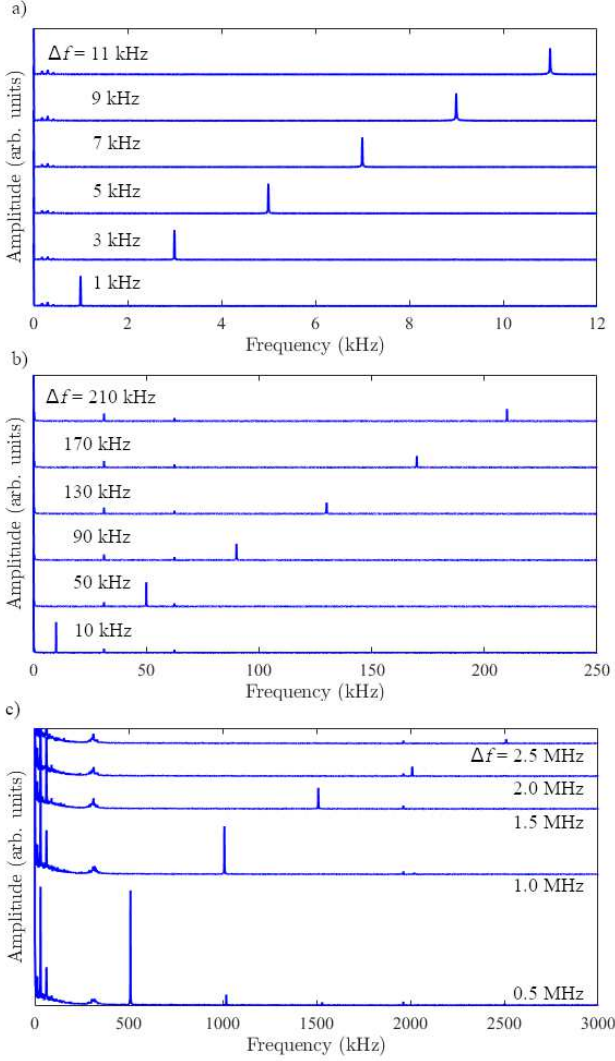


FIG. 3. DFT of the probe transmission for different values of Δf , keeping the LO frequency fixed. Each plot a) - c) has a different y-scale. a) shows values of Δf from bottom to top of 1 kHz, 3 kHz, 5 kHz, 7 kHz, 9 kHz and 11 kHz, b) shows values of 10 kHz, 50 kHz, 90 kHz, 130 kHz, 170 kHz and 210 kHz, and c) shows values of 500 kHz, 1 MHz, 1.5 MHz, 2 MHz, and 2.5 MHz.

19.628500 GHz. Each subfigure shows a different frequency range. The Δf signal gets weaker the further the frequency of SIG is detuned from the frequency of SIG (the different frequency ranges were plotted on different scales for illustration; the magnitude of the low-frequency noise gives some indication of the change in scale). The detuning is limited by the response rate of the EIT/AT effect. The minimum time for the EIT/AT to develop is on the order of $\sim 0.1 \mu s$ ^{12,13,17,19}, which limits the modulation frequencies that are observable to $\Delta f \lesssim 10$ MHz.

From eq. (7), we see that if we measure the phase ($\Delta\phi$) of the intermediate frequency signal in the probe transmission, we directly measure the phase between LO and SIG. A direct method for varying the phase of an RF field is to physically

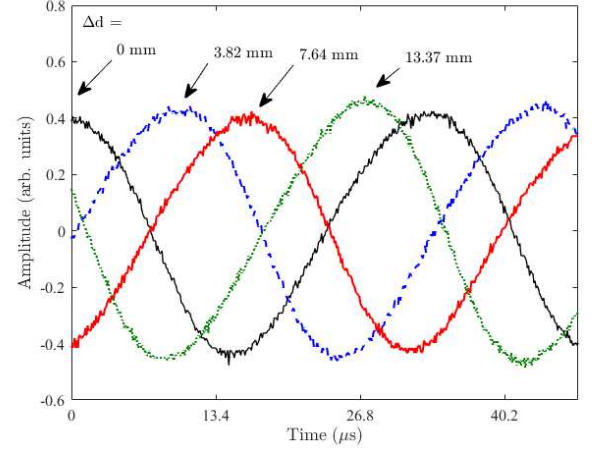


FIG. 4. Signal detected on photodetector for different SIG horn distances Δd . The IF frequency is $\Delta f = 30$ kHz.

translate the source. We demonstrate the ability of the Rydberg atoms to detect RF phase by translating the SIG horn linearly away from the atoms (in the z-direction), as shown in Fig. 2. The SIG horn is translated by a motorized linear stage, with a step size of $\Delta d = \lambda_{RF}/8 = 1.91$ mm. The LO frequency is set to $f_{LO} = 19.626000$ GHz, and the SIG is set at $f_{SIG} = 19.626030$ GHz, for a difference in frequency of $\Delta f = 30$ kHz. A function generator is set to 30 kHz for a reference signal to serve as the trigger for the oscilloscope. The two signal generators and the function generator are connected to a Rb clock 10 MHz signal. The LO horn antenna generates an electric field of $|E| = 2.97$ V/m at the vapor cell, and the SIG horn a field of $|E| = 1.22$ V/m.

By translating the SIG horn antenna while leaving the LO horn antenna fixed, we shift the relative phase between the SIG and LO fields at the vapor cell. We measure the change in phase ($\Delta\phi$) of our detected signal as a function of the distance traveled by the RF horn antenna Δd , giving us the propagation constant²⁰ β :

$$\beta = \frac{\Delta\phi}{\Delta d} . \quad (8)$$

Four sample traces corresponding to different distances of the SIG horn relative to the Rydberg atoms are shown in Fig. 4. We track the phase of the signal relative to the first trace as the SIG horn is moved, keeping track of the accumulated phase. Fig. 5 shows the measured phase difference vs. distance. This measurement was repeated 5 times and averaged. The fit to the data in Fig. 5 gives a $\beta = 409.3$ rad/m. The theoretical value for propagation constant for a wave propagating in free space is given by²⁰ $\beta = \omega_{RF}/c$. In this experimental setup (Fig. 2), the horn antenna for the SIG is at an angle of $\theta \approx 5^\circ$ from the orthogonal of the two laser beams. As such, the propagation constant measured in this setup must be corrected²⁰ by $\cos(\theta)$, giving:

$$\beta = \frac{\omega_{SIG}}{c} \cos(\theta) = 409.7 \frac{\text{rad}}{\text{m}} . \quad (9)$$

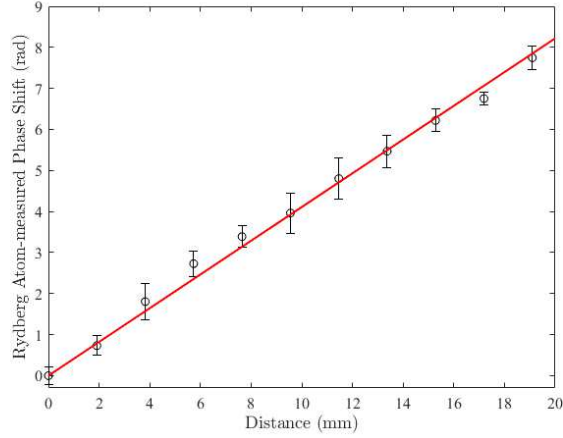


FIG. 5. Measured phase shift vs. distance SIG horn was moved. Errorbars are the standard deviation from 5 measurements. Line shows predicted phase shift, with slope from eq. (9) of 409.7 rad/m. The fit to the data gives a measured slope of 409.3 rad/m.

After correcting for the angle, the measured propagation constant is within 0.1% of the theoretical value. This shows that the phase of the SIG field was measured relative to the phase of LO.

This work demonstrates a Rydberg atom-based RF mixer, and the ability of Rydberg atoms to directly measure the phase of an RF field. The Rydberg atoms demodulate the RF field, and transfer the IF frequency directly to an optical field. The presence of a second RF field (analogous to an LO) creates a beat note that the Rydberg atoms can demodulate. The phase of the demodulated signal in the optical field is directly related to the phase of the incident RF field. The uncertainties of this type of phase measurement technique are currently being investigated. However, the results in this paper show that the phase can be determined to at least 0.1 % when compared to theoretical values. The ability to measure phase allows a single quantum sensor to fully characterize an RF field, at frequencies from MHz to THz. This allows atomic receivers to be competitive with current technologies and opens up a wide range of possible applications for Rydberg atom-based sensors. The Rydberg atom-based mixer can be used to open up several different avenues for quantum-based RF field detection, including communications applications, near-field antenna measurements, and radar. It is not necessary for the incident RF source to be physically connected to the LO source as long as the phases can be synchronized, for instance through

the use of atomic clocks. This would be necessary to use Rydberg atom receivers in high-frequency communications. Rydberg atoms have been shown to measure RF E-field amplitude^{1–6} and polarization⁷. This work shows that Rydberg atoms can also be used to measure RF phase, previously a missing link for complete E-field metrology with Rydberg atom-based quantum sensors.

- ¹C. Holloway, J. Gordon, S. Jefferts, A. Schwarzkopf, D. Anderson, S. Miller, N. Thaicharoen, and G. Raithel, *Antennas and Propagation, IEEE Transactions on* **62**, 6169 (2014).
- ²M. Tanasittikosol, J. D. Pritchard, D. Maxwell, A. Gauguier, K. J. Weatherill, R. M. Potvliege, and C. S. Adams, *Journal of Physics B: Atomic, Molecular and Optical Physics* **44**, 184020 (2011).
- ³J. Sedlacek, A. Schwettmann, H. Kubler, R. Low, T. Pfau, and J. Shaffer, *Nature Physics* **8**, 819 (2012).
- ⁴J. A. Gordon, C. L. Holloway, A. Schwarzkopf, D. A. Anderson, S. Miller, N. Thaicharoen, and G. Raithel, *Applied Physics Letters* **105**, 024104 (2014), <http://dx.doi.org/10.1063/1.4890094>.
- ⁵C. L. Holloway, M. T. Simons, J. A. Gordon, P. F. Wilson, C. M. Cooke, D. A. Anderson, and G. Raithel, *IEEE Transactions on Electromagnetic Compatibility* **59**, 717 (2017).
- ⁶H. Fan, S. Kumar, J. Sedlacek, H. Kübler, S. Karimkashi, and J. Shaffer, *J. Phys. B: At. Mol. Opt. Phys.* **48**, 202001 (2015).
- ⁷J. A. Sedlacek, A. Schwettmann, H. Kübler, and J. P. Shaffer, *Phys. Rev. Lett.* **111**, 063001 (2013).
- ⁸T. F. Gallagher, *Rydberg Atoms*, Cambridge Monographs on Atomic, Molecular and Chemical Physics (Cambridge University Press, 1994).
- ⁹A. K. Mohapatra, T. R. Jackson, and C. S. Adams, *Phys. Rev. Lett.* **98**, 113003 (2007).
- ¹⁰M. T. Simons, J. A. Gordon, C. L. Holloway, D. A. Anderson, S. A. Miller, and G. Raithel, *Applied Physics Letters* **108**, 174101 (2016), <http://dx.doi.org/10.1063/1.4947231>.
- ¹¹D. Stack, B. Rodenburg, S. Pappas, W. Su, M. S. John, P. Kunz, M. Simons, J. Gordon, and C. Holloway, in *APS DAMOP* (2017).
- ¹²D. Meyer, K. Cox, F. Fatemi, and P. Kunz, *Applied Physics Letters* **12**, 211108 (2018).
- ¹³K. C. Cox, D. H. Meyer, F. K. Fatemi, and P. D. Kunz, *Phys. Rev. Lett.* **121**, 110502 (2018).
- ¹⁴Z. Song, W. Zhang, H. Liu, X. Liu, H. Zou, J. Zhang, and J. Qu, [arXiv:1808.10839 \[physics.atom-ph\]](https://arxiv.org/abs/1808.10839) (2018).
- ¹⁵D. Anderson, R. E. Sapiro, and G. Raithel, [arXiv:1808.08589 \[physics.atom-ph\]](https://arxiv.org/abs/1808.08589) (2018).
- ¹⁶“Get ready for atom radio,” (MIT Technology Review, 2018) <https://www.technologyreview.com/s/611977/get-ready-for-atomicradio/>.
- ¹⁷C. L. Holloway, M. T. Simons, A. H. Haddab, J. A. Gordon, and S. D. Vorum, “A multiple-band rydberg-atom based receiver/antenna: Am/fm stereo reception,” (2019), *IEEE Antennas and Propagation Magazine* (submitted).
- ¹⁸M. T. Simons, A. H. Haddab, R. Horansky, and C. L. Holloway, “Atom-based receiver: a pseudo-random bit sequence in the presence of noise,” (2019), *IEEE Antennas and Propagation Letters* (in progress).
- ¹⁹Y. qing Li and M. Xiao, *Opt. Lett.* **20**, 1489 (1995).
- ²⁰C. A. Balanis, “Advanced engineering electromagnetics,” (Wiley, 1989) Chap. 4.